

PS 3: Week 6

Quantifying Uncertainty

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Agenda

- ① Bias vs. Noise
- ② Standard Errors
- ③ P-Values
- ④ Practice Exercise

Reminders

- Problem Set 1 due today (Friday) at midnight
 - Verify you submitted the right assignment. I have started grading some and some of them are a in-class assignment
- Project Proposal due Monday at midnight
- Midterm signup will go out Monday morning

Gradescope Error

If you ever see this error on a gradescope submission, do not hesitate to email. It is because the autograder failed for some reason and I can have it rerun.

Autograder Results

[Results](#)[Code](#)

The autograder failed to respond in the expected amount of time. If the autograder continues to fail, please contact help@gradescope.com.

The autograder failed to execute correctly. Contact your course staff for help in debugging this issue. Make sure to include a link to this page so that they can help you most effectively.

Research Proposal Feedback

Is the research question clear?

Is it easy to tell if it is a causal/descriptive question?

Is the data you are planning to collect well suited to answer the question?

Measuring Uncertainty

Discuss:

- 1 Why do we care about measuring the uncertainty of an estimate? In other words, why just the difference in means not enough to make conclusions about the effect of a treatment?
- 2 What is the difference between noise and bias?

SAMPLING BIAS



"WE RECEIVED 500 RESPONSES AND
FOUND THAT PEOPLE LOVE RESPONDING
TO SURVEYS"

sketchplanations

- Bias is a systematic deviation from the truth
- You can think of bias as a consistent inaccuracy
- You've already seen bias!
- Similar to bias in a colloquial sense:
 - If the judicial system is racially biased, it will be consistently discriminatory
 - If I have a biased point of view, I will automatically come to the same conclusion regardless of the evidence you present to me
 - If we have a biased way of researching our topic of interest, we will end up at the same (wrong) conclusion

- Noise is simply the random difference between samples
- Noise is inevitable when sampling – we don't have the true population

Bias vs. Noise

Bias can also be thought of as Inaccuracy

Noise can be thought of as Imprecision



Low Bias
Low Noise



Low Bias
High Noise



High Bias
Low Noise



High Bias
High Noise

1. Standard Deviation

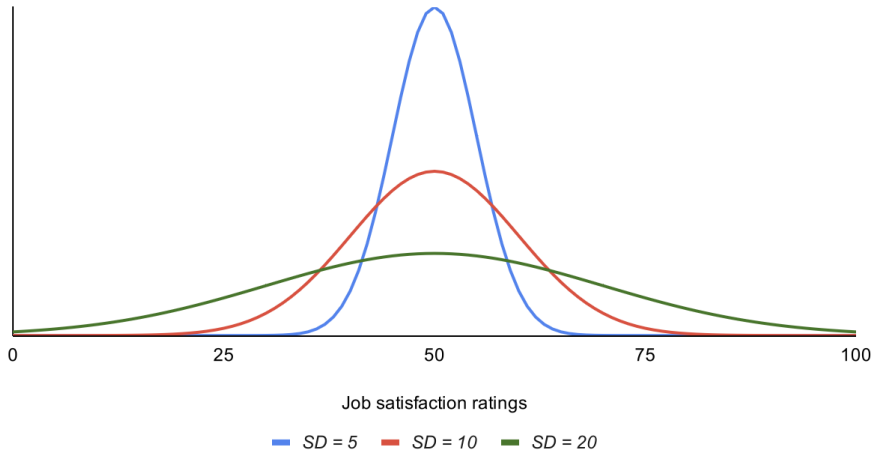
A measure of “spread” – summary of how much a set of values are away from the mean

The formula is:

$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Standard Deviation

Job satisfaction ratings of three groups



2. Standard Error

- If we took a bunch of samples and measured a value for each, the standard error is the standard deviation of those values.
- In other words, standard error is a measure of how much we'd expect our findings to vary if we could replicate our study thousands of times
- Therefore, the standard error helps us measure how much we'd expect the estimate from the experiment to differ from the truth (the real effect) by chance
- It is a measure of noise, not bias

Standard Error and other sample statistics

$$\text{Standard Error} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- n = Sample size (i.e., number of observations)
- s = Sample standard deviation
- $s_{\bar{x}}$ = Estimated standard error of the mean ($s/\sqrt{n}$)

Standard Error other sample statistics

$$\text{Standard Error} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- It can only be positive
- What do you think happens to the SE as s or n increases?

Standard Error other sample statistics

$$\text{Standard Error} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

- It can only be positive
- What do you think happens to the SE as s or n increases?

As s increases, the standard error will increase as well. As the spread of the population gets bigger (standard deviation), the actual sample that we get from it will also grow.

3. t-statistic

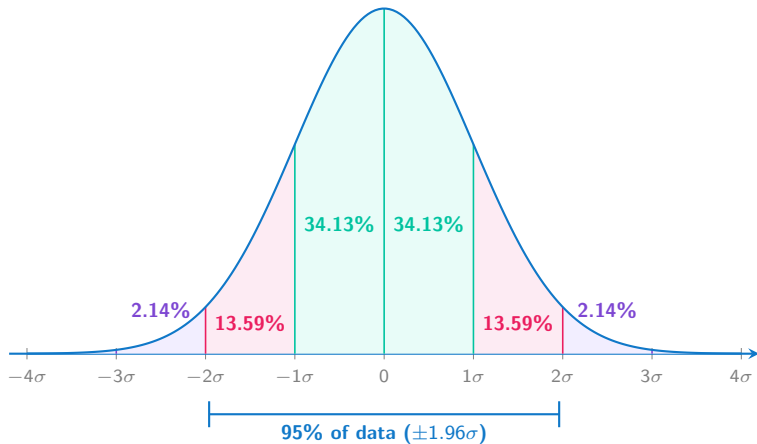
$$t = \frac{\text{Estimate}}{\text{Standard Error}}$$

- **t-statistic:** to measure how likely that an estimate of the size we saw would arise by chance, even if the treatment had no effect.
- **Estimate:** our best guess of what the true ATE is.
- If $t > 1.96$ or $t < -1.96$, the observed result is **statistically significant** (more next week).

Design: Standard

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper
treatsociallobby	0.120122	0.07553198	1.590346	0.1162785	-0.03053066	0.2707746
		DF				
treatsociallobby	69.76772					

t-statistic: Why 1.96?



}

Interpreting T-Statistics

Design: Standard

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
treat	0.01857536	0.01099631	1.689236	0.091361	-0.002992649	0.04014337	1669.581

$$0.01857536 / 0.01099631 = 1.689236$$

A small estimate relative to a large standard error will result in a small t-statistic (and less confidence in the estimate being different from zero).

Hypothesis Testing

- In science we test hypotheses
 - Ex: Autocracies are more likely than democracies to go to war
- The **null hypothesis** is our default: what the world would look like if we did not observe the effect we are interested in finding
 - Ex: Autocracies and democracies are equally likely to go to war
- The **alternative hypothesis** reflects what we would observe if there was, actually, a true effect in the world
 - Ex: Autocracies and democracies are not equally likely to go to war
- We never prove one hypothesis or the other
 - Because of noise (i.e. sampling), we can never be sure we're observing the truth
 - But we can measure how likely we'd be to observe the effect we see by chance

- What's the probability that we would observe the effect we see by chance?
 - A p-value of 0.13 means that there is a 13% probability that we would observe the effect we see even if the null hypothesis was true
- We want to maximize our confidence that we are NOT observing the effect by chance
 - A p-value of greater than (or equal to) 0.05 is considered statistically insignificant
 - A p-value of less than 0.05 is statistically significant
 - We are less than 5% likely to observe this effect by chance!
 - OR: We are more than 95% likely to not be observing this effect by chance!
- This gives us evidence in support of the alternative hypothesis but does not prove it

Interpreting P-Values

Design: Standard

	Estimate	Std. Error	t value	Pr(> t)	CI Lower	CI Upper	DF
treat	0.01857536	0.01099631	1.689236	0.091361	-0.002992649	0.04014337	1669.581

- The treatment increases the outcome by about 1.9 percentage-points relative to the control.
- However, we would expect this result (or something smaller) to occur roughly 9% of the time if this effect was driven entirely by chance.

Attendance Survey



I have created a practice exercise for you to work through to give you more practice with R and a chance to determine the steps yourself and then implement it.

It is intended to be harder than the in-class assignments so don't be discouraged.

Go to the bCourses page → Section 6 → Week 6 Section Exercise

I also have created *optional* exercises that will teach you more R beyond what the class will cover. If you would prefer, you can also use this time to work on that.

Acknowledgements

Special thanks to prior GSIs, Will Halm, and Tess Louie for providing materials off of which some of these slides are based.

- We want to show that our findings are not due to chance
 - This arises from concerns about noise
 - Different sampling methods and differences in randomization could lead to slightly different results
- Standard errors show how much our results would vary across multiple versions of the same study
 - How much “error” (difference) in our outcome should we expect across samples?
- We test hypotheses to provide support for one vs. another
 - We never provide definitive proof for any hypothesis!
 - But p-values show us how probable our results would be if they arose due to chance and not due to a treatment effect

An Interpretation

Imagine an experiment with two treatments, Treatment 1 and Treatment 2. The outcome is a binary measure of vote choice: whether one votes for Party A. Participants are randomly assigned to T1, T2, or the control (no treatment).

40% of the control group votes for Party A, 45% of those in T1 vote for Party A, and 50% of those in T2 vote for Party A.

My interpretation: Participants under Treatment 1 were 5-percentage points more likely to vote for Party A than those in the control. Participants under Treatment 2 were 10-percentage points more likely to vote for Party A than the control. These results prove that both treatments would increase vote choice for Party A among the general population. Treatment 2 would also be the more effective treatment at increasing Party A's vote choice if we ran this study again.

What's wrong with this interpretation? What's correct about it?

What Do We Know?

- In this sample, Treatment 1 increased vote for Party A by 5-percentage points relative to the control
- In this sample, Treatment 2 increased vote for Party A by 10-percentage points relative to the control
- In this sample, Treatment 2 increased vote for Party A by 5-percentage points more than Treatment 1 did relative to the control
- In short, we know we have an average treatment effect, but we don't know the true treatment effect
 - We never see everyone's outcomes on both treatments and control! Just on one assignment!
 - So we sampled participants and then randomly assigned treatment/control within our sample
 - What if we had sampled different participants? What if we had assigned treatment differently?
- We need a way to measure how confident we are that the findings in this study represent a true average treatment effect.
 - We can never prove that they do, but we can increase our confidence that what we found is not a coincidence.

The 1936 Election: Landon vs Roosevelt

- Literary Digest did one of the largest political polls (at the time) asking 2.4 million people who would win the 1936 presidential election
 - Their prediction: Landon = 57% vs Roosevelt = 43%
- Their method: Ask 10 million people on telephone lists, magazine subscription lists, club and association rosters, to fill out a mock ballot

In reality: Landon = 38% vs Roosevelt = 62%

